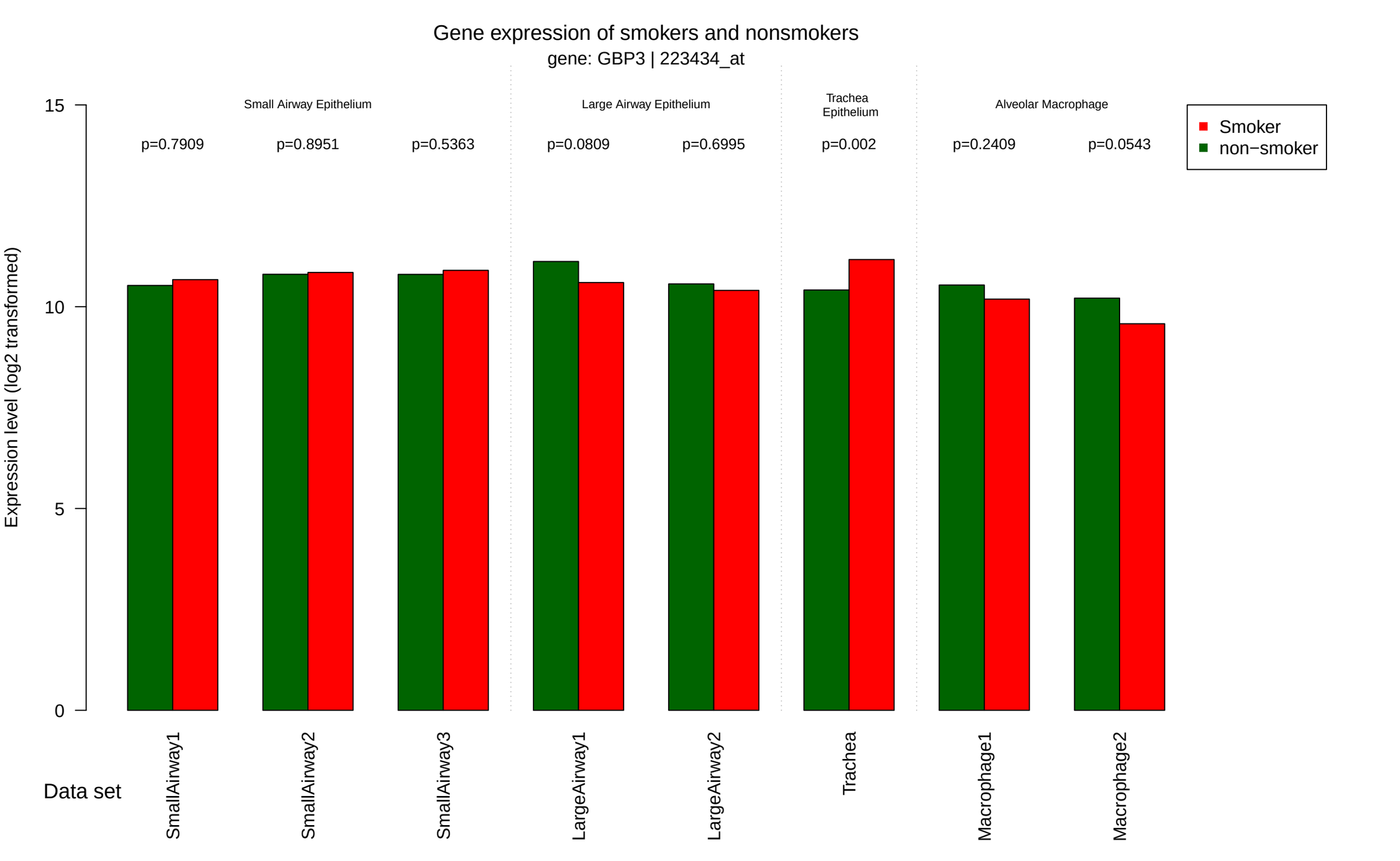
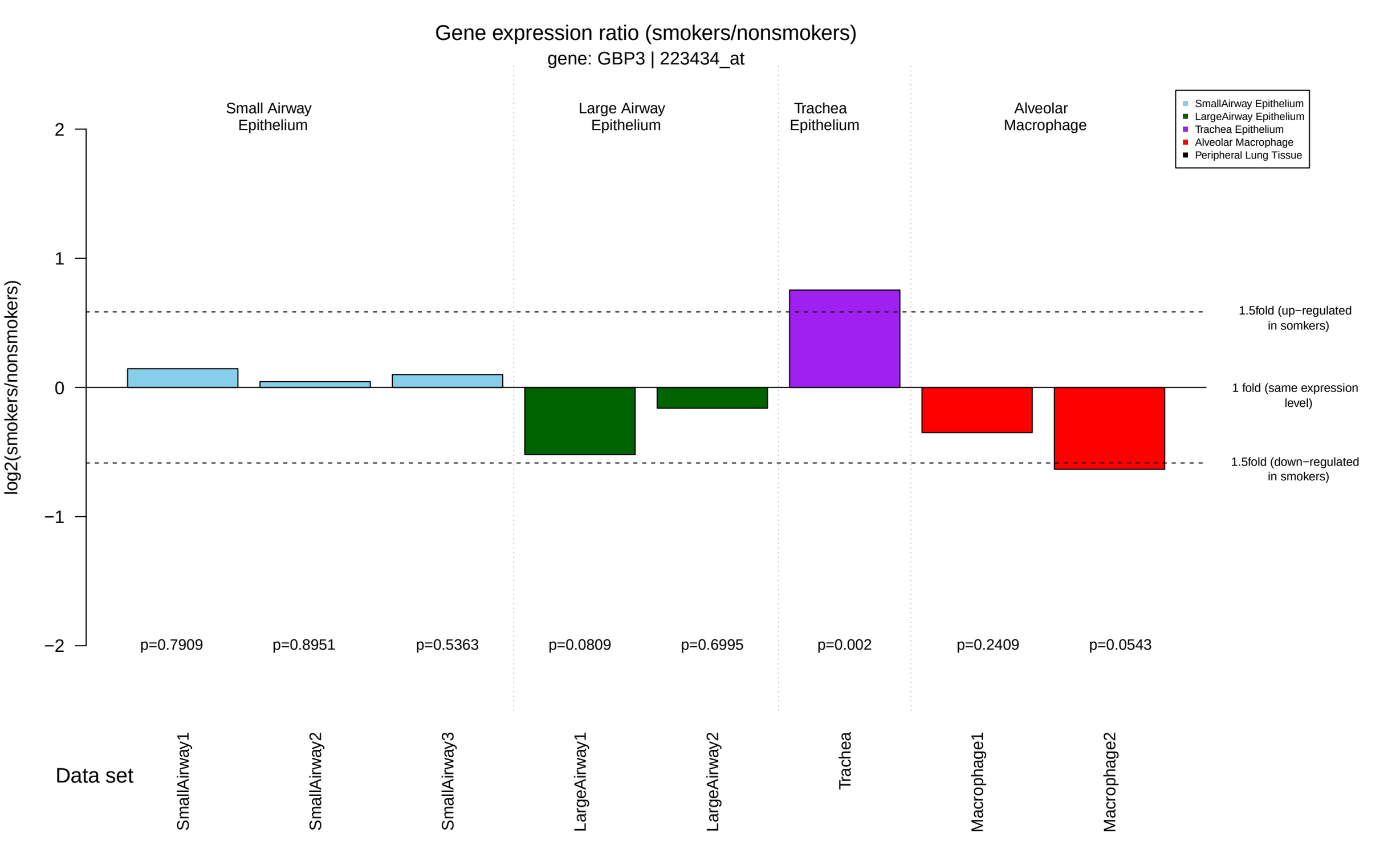
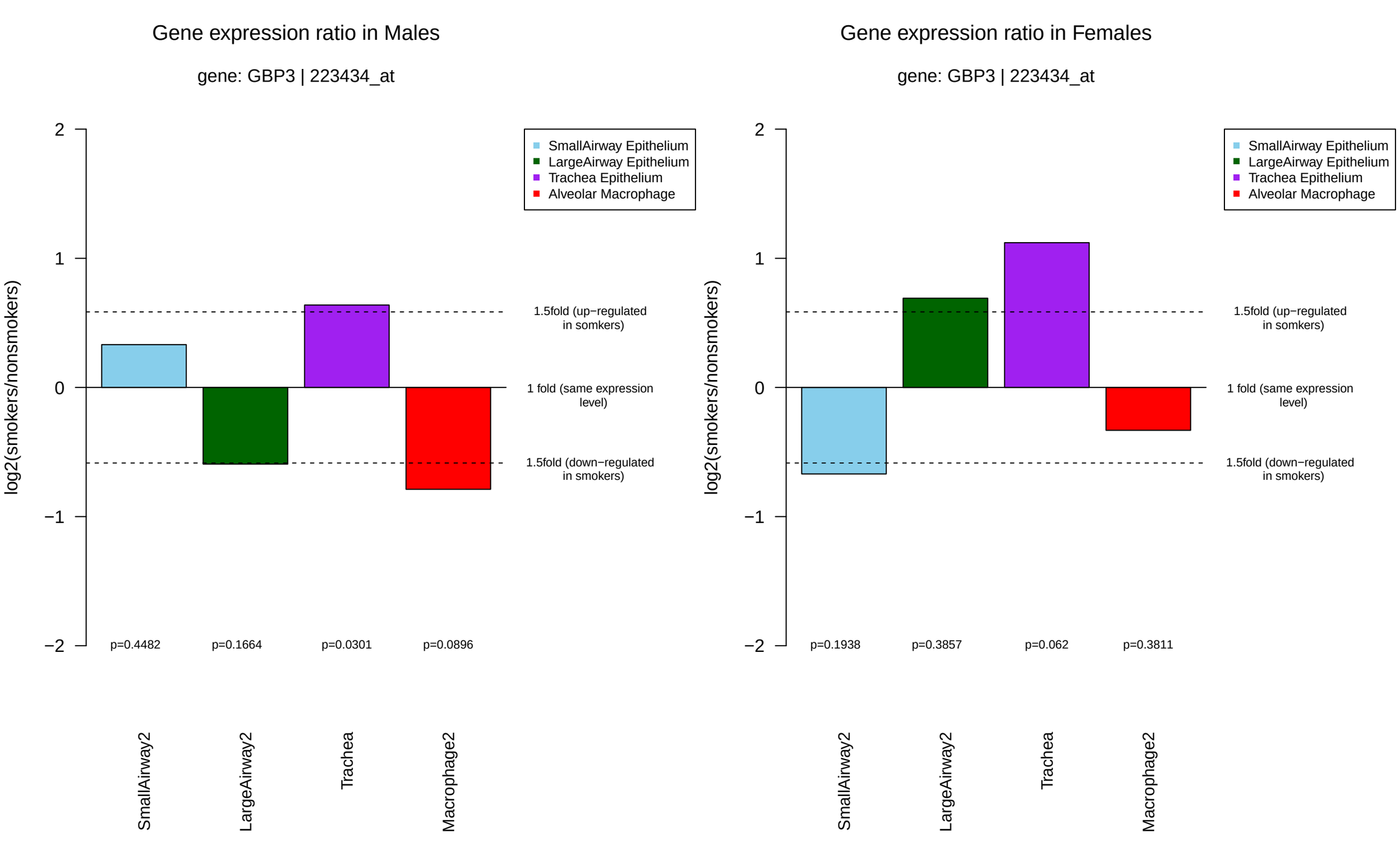
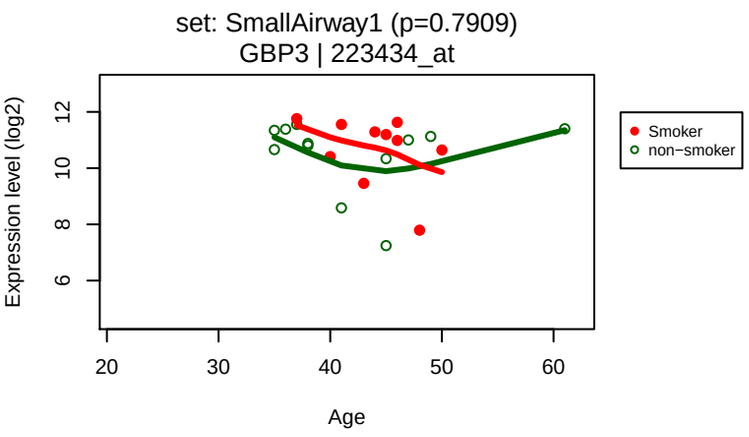
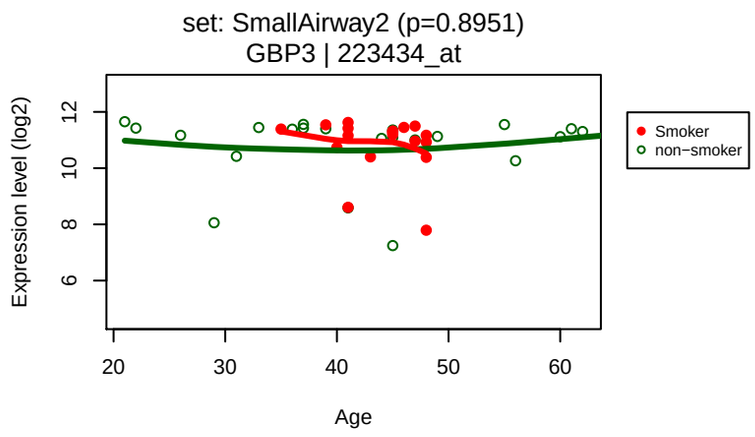
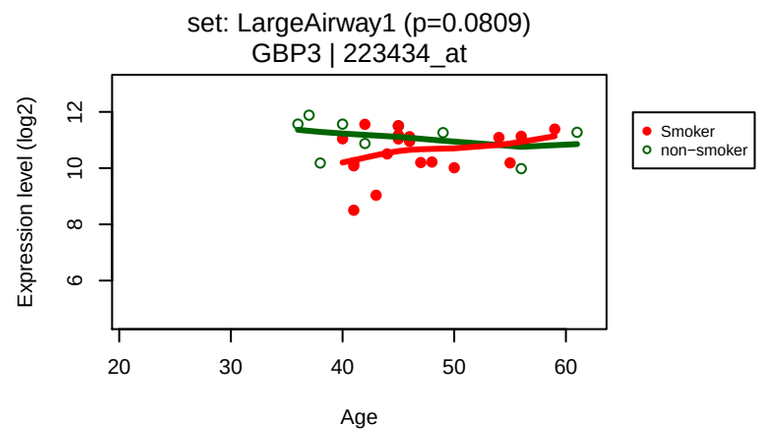

Gene expression of smokers and nonsmokers
gene: GBP3 | 223434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7909
p=0.8951
p=0.5363
p=0.0809
p=0.6995
p=0.002
p=0.2409
p=0.0543
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

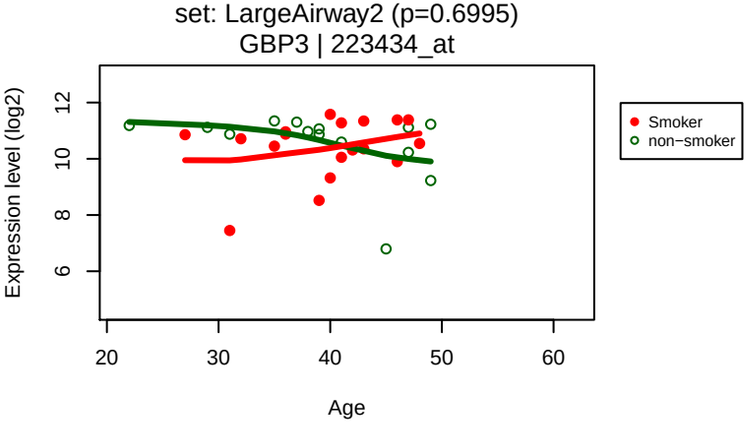

Gene expression ratio (smokers/nonsmokers)
gene: GBP3 | 223434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7909
p=0.8951
p=0.5363
p=0.0809
p=0.6995
p=0.002
p=0.2409
p=0.0543
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

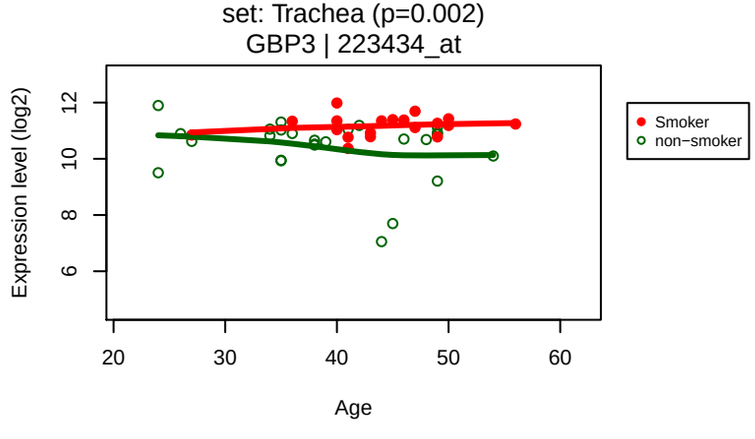

Gene expression ratio in Males
gene: GBP3 | 223434_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4482
p=0.1664
p=0.0301
p=0.0896
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GBP3 | 223434_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1938
p=0.3857
p=0.062
p=0.3811
SmallAirway2
LargeAirway2
Trachea
Macrophage2

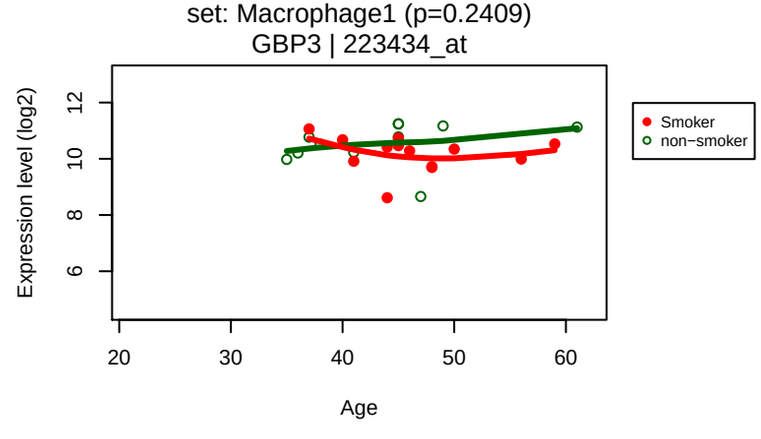

set: SmallAirway1 (p=0.7909)
GBP3 | 223434_at
Expression level (log2)
Age
Smoker
non−smoker

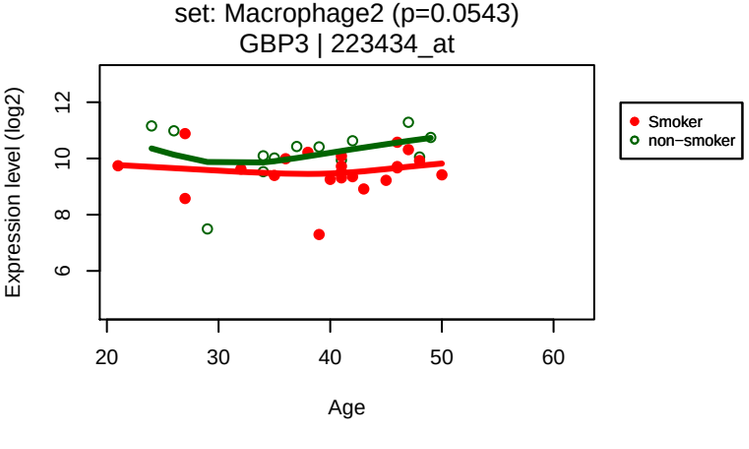

set: SmallAirway2 (p=0.8951)
GBP3 | 223434_at
Expression level (log2)
Age
Smoker
non−smoker

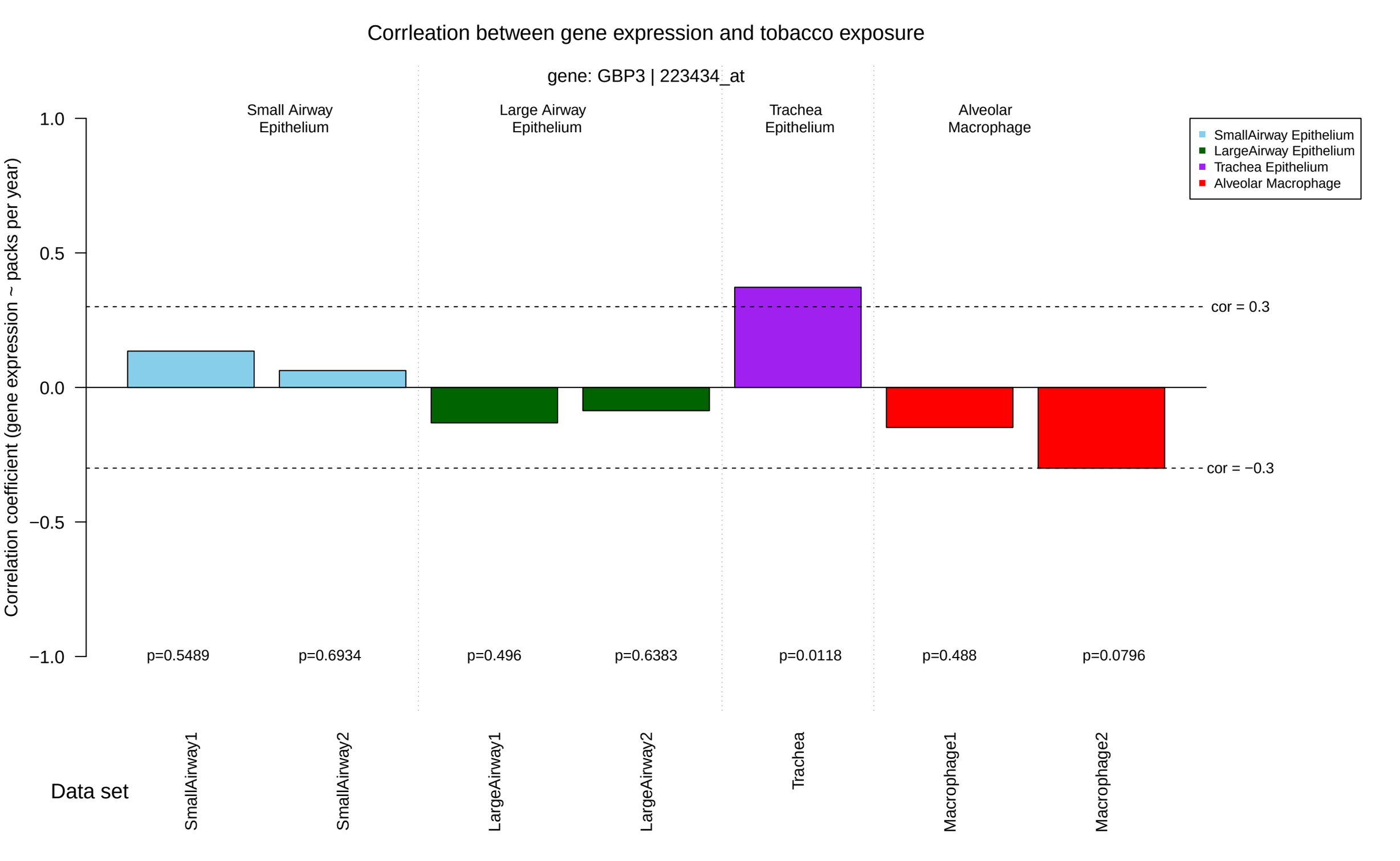

set: LargeAirway1 (p=0.0809)
GBP3 | 223434_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6995)
GBP3 | 223434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.002)
GBP3 | 223434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2409)
GBP3 | 223434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0543)
GBP3 | 223434_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GBP3 | 223434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5489
p=0.6934
p=0.496
p=0.6383
p=0.0118
p=0.488
p=0.0796
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2